

LEXREG AFRICA

Development Audit & Implementation Report



Week of Development Review

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EXECUTIVE SUMMARY

This document presents a comprehensive audit of the LexReg Africa platform codebase, detailing all implemented features, architectural decisions, and the current state of development. The platform is built as a Next.js 16 application with a modern React 19 frontend, Supabase backend, and Sanity CMS for content management.

The implementation to date covers the full public-facing landing experience, a robust authentication system, onboarding path selection, a content-managed insights blog, and the foundational dashboard architecture. The design system is fully operational, grounded in an iOS 18 / Apple Business Design System philosophy with a bespoke Navy & Gold brand palette.

Module	Status	Notes
Landing Page	COMPLETE	All sections implemented with animations, 3D globe, and interactive bento grid
Authentication	COMPLETE	Login, signup, forgot password with Supabase Auth & cookie-based SSR
Onboarding	PARTIAL	Path selection screen complete; entity-specific flows pending
Dashboard	SCAFFOLD	Layout shell exists; functional widgets pending
CMS / Insights	COMPLETE	Sanity Studio + insights blog with placeholder fallback data
Design System	COMPLETE	Full CSS tokens, iOS typography scale, 30+ custom UI components
Database	COMPLETE	Supabase schema generated with TypeScript types; 20+ tables defined

1. TECHNOLOGY STACK & ARCHITECTURE

The LexReg Africa platform is architected as a modern full-stack application with clear separation between presentation, business logic, and data layers.

1.1 Core Technologies

Layer	Technology	Version	Purpose
Framework	Next.js	16.2.9	App Router, SSR, API routes, font optimization
Runtime	React	19.2.4	UI component model, concurrent features
Language	TypeScript	5.x	Type-safe development across full stack
Styling	Tailwind CSS	4.x	Utility-first CSS with custom design tokens
UI Library	shadcn/ui	4.11.0	Accessible, composable component primitives
Animation	Motion (Framer Motion)	12.42.0	Scroll-driven, gesture, and layout animations
State	Zustand	5.0.14	Lightweight global state management
Backend	Supabase	2.108.2	Auth, PostgreSQL DB, RLS, storage, Edge Functions
CMS	Sanity	6.2.0	Structured content management for insights/blog
Icons	Tabler Icons	3.44.0	Consistent, lightweight SVG icon system
3D / WebGL	Three.js + React Three Fiber	0.185.0 / 9.6.1	Interactive 3D globe visualization

2. IMPLEMENTED FEATURES — DETAILED BREAKDOWN

2.1 Public Landing Experience

The landing page is a fully interactive, animation-rich single-page experience designed to communicate trust, professionalism, and regulatory expertise. It comprises seven distinct sections with a cohesive visual narrative:

- **Hero Section** — Cloudscape WebGL shader background with animated sky gradient, KineticText hover-reactive headline ('Start strong. Stay compliant.'), ShimmerButton CTA with gold shimmer effect, and frosted-glass secondary CTA.
- **Trust Bar** — LogoCloud grid displaying all 8 Kenyan regulatory bodies (BRS, KRA, CAK, CMA, CBK, KEBS, ODPC, NSSF) with the 'Designed around Kenya's regulatory framework' copy.
- **How It Works** — Scroll-driven TextReveal headline ('From idea to fully compliant entity') followed by an interactive 5-card bento grid with animated data visualizations: Registration Pipeline (SVG flow diagram), Compliance Score (animated bar charts), Task Checklist (stacked carousel), Document Vault (namespace bars + activity log), and Service Pillars (category grid).
- **Services Section** — Three-pillar expandable layout with navigation arrows, animated pill tags, and contextual right-panel visuals: Entity Formation pipeline, Corporate Services document list, and Legal Audit checklist with progress bar.
- **Globe Section** — 3D interactive globe (Three.js + three-globe) centered on Nairobi with animated gold arcs connecting Kenya to East Africa, wider Africa, Middle East, Europe, Asia, and the Americas. Auto-rotate with atmosphere glow.
- **Insights Section** — Bento grid of 5 content cards pulling from Sanity CMS with featured article, 2 small cards, 1 tinted card, and 1 wide card. Includes category badges, read-time indicators, and hover scaling.
- **Footer** — 'Taped' card design with decorative tape SVG elements, three-column navigation (Services, Platform, Company), social links (LinkedIn, X), and legal links.

2.2 Navigation & Global UI

The NotchNavbar is a custom-built fixed navigation component with a distinctive 'notch' silhouette inspired by premium hardware design. It features frosted glass (backdrop-filter) surfaces, smooth scroll anchor links, mobile hamburger menu with AnimatePresence dropdown, and ShimmerButton CTAs. The CanvasFractalGrid provides a fixed golden-dot background with wave-distortion mouse tracking, gradient animations, and noise overlay across all pages.

2.3 Authentication System

Complete email/password authentication via Supabase Auth with SSR-safe cookie management. The auth callback route handles both OAuth code exchange and OTP verification (email confirmation, password reset, invite flows). Forms follow iOS design patterns with grouped input rows, system label colors, and hairline separators.

- **Login** — Email + password with grouped inputs, error handling, loading states, and 'Create Account' secondary CTA.

- **Signup** — Full name, email, password (8+ char minimum) with terms acknowledgment, loading states, and post-signup redirect to onboarding.
- **Forgot Password** — Email input with reset link dispatch, success state with mail icon, and 'Back to Sign In' navigation.

2.4 Onboarding Path Selection

The onboarding page presents three distinct paths in a card-based layout, each with a category icon, label, sublabel badge, description, and chevron navigation. This is the first step of the entity onboarding journey after authentication.

- **Existing Business** (BRS registered) — Document upload and OCR verification path.
- **Register New Entity** (Not registered) — Guided questionnaire through BRS registration.
- **Informal Business** (Informal) — Readiness assessment with personalized formalization path.

2.5 Content Management (Sanity CMS)

Sanity is configured as the headless CMS for the insights/blog engine. The schema defines an 'insight' document type with full editorial capabilities: title, slug (auto-generated from title), summary (max 200 chars), category (6 options: Compliance, Registration, Governance, Tax, Legal, Data & Privacy), cover image with hotspot, featured boolean, read time, publish date, and full Portable Text body with block and image support.

The CMS includes a custom Sanity Studio embedded at /studio using next-sanity, with structure tool plugin for content management. The public insights page uses ISR with 60-second revalidation, placeholder fallback data for offline development, and generateStaticParams for static generation at build time.

2.6 Database Schema (Supabase)

The Supabase PostgreSQL schema has been fully typed with 1,076 lines of generated TypeScript types. The schema covers the full multi-tenant governance platform with 20+ tables including:

- **audit_logs** — Immutable action logs with IP address, metadata, organisation linkage, and resource tracking.
- **organisations** — Entity profiles with type, registration number, KRA PIN, tax status, and soft-delete support.
- **profiles** — User profiles linked to Supabase Auth with role, phone, and avatar support.
- **compliance_events** — Regulatory deadline tracking with recurrence, status workflow, and notification triggers.
- **documents** — Document vault with versioning, namespace tagging, OCR metadata, and entity linkage.
- **directors, shareholders, trustees** — Stakeholder management with KYC fields, shareholding, and appointment dates.

2.7 Design System & Component Library

A comprehensive design system has been implemented with 30+ custom components and a full CSS custom property token layer. Key components include:

- **ShimmerButton** — Conic-gradient shimmer animation with configurable background, shimmer color, and duration.
- **ShinyButton** — Mask-driven sweep animation with spring physics and tap scaling.
- **KineticText** — Per-character hover weight transition (300→900) with stroke width animation.
- **TextReveal** — Scroll-driven word-by-word opacity reveal over 200vh scroll distance.
- **TextAnimate** — Blur-in-up word animation with staggered timing and configurable delay.
- **CanvasFractalGrid** — High-performance Canvas 2D dot grid with mouse wave distortion, gradient animation, and adaptive quality (low/medium/high FPS).
- **Cloudscape** — Custom WebGL fragment shader with FBM noise, wind morphing, and three-color band mixing.
- **Highlighter** — Rough-notation highlight wrapper with configurable action, color, stroke width, and view-trigger.

- **LexRegLogoMark** — Custom SVG logo with Africa continent outline and gold swoosh, parameterized by size and color.

3. PROJECT STRUCTURE

The codebase follows Next.js App Router conventions with clear separation of concerns:

Path	Contents
app/	Next.js App Router: layout.tsx, page.tsx, route handlers, route groups (auth, dashboard, onboarding), dynamic insights/[slug]
components/landing/	7 section components: hero-section, how-it-works-section, services-section, globe-section, insights-section, footer-section, insights-bento-grid
components/ui/	20+ reusable UI primitives: buttons, navbar, logo, text effects, grid, form elements, highlighter
components/auth/	LoginForm, SignupForm, ForgotPasswordForm with Supabase client integration
components/forgeui/	Cloudscape WebGL shader component for hero background
lib/	Utility libraries: supabase server/client, sanity client/queries, cn() helper
sanity/	Schema definitions (insight.ts), schemaTypes index, CLI and studio config
types/	Generated Supabase database.types.ts (1,076 lines)

4. PENDING IMPLEMENTATION & NEXT STEPS

The following items represent the next phase of development required to reach MVP readiness:

4.1 Dashboard (High Priority)

- Entity dashboard with quick stats, compliance score widget, and upcoming deadlines
- Document vault tab with upload, search, versioning, and namespace filtering
- Compliance calendar with recurring event creation, status workflow, and email reminders
- Stakeholder management (directors, shareholders, trustees) with KYC document linking

4.2 Onboarding Flows (High Priority)

- Existing Entity: Multi-step document upload with OCR preview, confidence threshold, and manual correction
- New Entity: BRS questionnaire with conditional logic per entity type, save-as-draft, and progress persistence
- Informal Business: 10-15 question readiness assessment with maturity scoring and gap analysis dashboard

4.3 Paid Service Pillars (Medium Priority)

- Entity Formation & Registration: Document generation, BRS filing helper, certificate upload and verification
- Corporate Services: LexReg Diligence™ auto-generation of board resolutions, AGM notices, and corporate registers with lawyer review workflow
- Legal Audit: Comprehensive health check across statutory, contractual, and regulatory obligations with written report delivery

4.4 Integrations (Medium Priority)

- Paystack payment integration for M-Pesa, card, and bank payments per service pillar
- Google Document AI for OCR extraction of certificates, CR12, and ID documents
- KRA verification mock (manual verification for MVP with API integration deferred)

4.5 Compliance & Legal (Pre-Launch)

- Privacy Policy and Terms of Service pages (drafted, pending Charles review)
- ODPC registration and Data Protection Officer training
- Row-level security policy hardening and penetration testing

5. RISK ASSESSMENT & MITIGATION

Risk	Severity	Mitigation
BRS has no public API — manual filing required	HIGH	Hybrid workflow: system generates documents, user/lawyer manually files, uploads certificate back
Solo developer capacity vs. enterprise scope	HIGH	Ruthless MVP scoping: one entity type, one onboarding path, basic dashboard first
OCR accuracy for Kenyan documents	MEDIUM	60% confidence threshold with human review; Google Document AI + fallback to manual
Payment integration complexity	MEDIUM	Paystack business account setup; per-service payment triggers with clear UX
Lawyer role ambiguity	MEDIUM	Clarify assignment logic: manual by Charles, auto by workload, or hybrid by tier
Data Protection Act compliance	MEDIUM	RLS enabled, 30-day soft delete, audit logs, privacy policy, ODPC registration pre-launch

6. CONCLUSION & RECOMMENDATIONS

The LexReg Africa platform has achieved significant technical milestones in the current development phase. The public-facing experience is production-ready in terms of design, animation, and content management. The authentication layer is solid and secure. The design system is comprehensive and will accelerate all future UI development.

The critical path to MVP now runs through: (1) dashboard functional widgets, (2) onboarding flow implementation with save-as-draft and progress persistence, (3) payment integration for the three service pillars, and (4) lawyer-mediated workflow definition.

Recommendation: Prioritize the Existing Entity onboarding path with document upload and basic dashboard display. This delivers the smallest usable product that demonstrates platform value while Charles finalizes scope for Corporate Services and Legal Audit pillars.



This report was generated automatically from codebase analysis. For questions, contact Ian Love Mangirwa at 27 June 2026.